

Social Buzz Content Analysis

Which content should Social Buzz make more of?

*A personal portfolio project, inspired by the Accenture "Social Buzz" case study (a Forage job simulation).
Built from scratch in Python -- this is an independent learning project, not an official Forage submission or certificate.*

Objective

Social Buzz is a made-up social media company used for this project. People post content there, and other people react to it -- not just with a simple "like," but with over 100 different reaction types (like "love," "scared," or "cherish"). Every reaction is worth a certain number of points.

Nobody at Social Buzz had studied this data before. The goal of this project was simple to state: find out which types of content (like "cooking" or "science") get the most positive reactions, so the company knows what to invest more time and money into. To answer that, the raw data had to be cleaned up, combined into one usable table, added up, and turned into charts that make the answer obvious at a glance.

Tools and Techniques Used

- Python -- the programming language used for the entire project.
- Pandas -- a tool that lets Python open, clean, and work with spreadsheet-style data (the same kind of data you would see in Excel).
- Matplotlib -- a tool for drawing clean, simple charts.
- Plotly -- a tool for drawing interactive charts you can click and hover over to see exact numbers.

Features Used

In plain terms, here is what each tool actually did in this project:

Pandas

- Opened each CSV file (a simple spreadsheet file) and took a first look at what was inside, including checking for missing information.
- Cleaned up messy text -- made everything lowercase and removed stray quote marks stuck to some category names.
- Removed rows that were missing important information, instead of guessing what they should say.
- Combined separate tables into one, matching rows up by a column they had in common (for example, matching every reaction to the post it belonged to).
- Removed a column that was no longer needed after combining the tables.
- Grouped matching rows together (for example, all "science" posts) and added up their scores.
- Sorted results from highest to lowest, and kept just the top few.
- Counted how many times each value appeared (for example, how many "positive" reactions there were).
- Converted plain text dates into real dates Python could actually understand and do math on.
- Used a small custom function (a mini tool written just for this project) to turn an hour of the day, like 14,

into an easy label like "Afternoon."

Matplotlib

- Drew bar charts and a pie chart, with a consistent, modern color scheme applied throughout.

Plotly

- Drew interactive bar charts you can hover over for exact numbers, styled to match a professional dashboard.

Challenges Faced

Missing Information

- Some rows in the reactions file were incomplete -- about 3,000 rows were missing who reacted, and about 1,000 were missing what type of reaction it was. These incomplete rows were removed rather than guessed at.

Messy, Inconsistent Text

- Category names (like "science" or "cooking") were written inconsistently -- sometimes capitalized, sometimes with stray quote marks stuck to the text. What should have been 16 clean categories showed up as 41 different-looking versions. This was fixed by lowercasing everything and stripping the stray quote marks.

Combining the Files Correctly

- Two of the files both had a column called "User ID" -- but they meant different people in each file (the person who posted vs. the person who reacted). Combining the files required carefully sorting out this mix-up so the wrong information never got blended together.

Confusing Date Format

- Dates in the file were written day-first (22-04-2025 means April 22nd), not the month-first style common in the US. Python had to be told this directly -- otherwise some dates could have been silently misread with the day and month swapped, with no error or warning.

A Computer Setting Blocked the Code

- Partway through, a Windows security feature blocked one of the tools from running at all, with no clear error message at first. This had to be diagnosed and turned off before the project could continue.

Methodology

Step 1: Clean the Data

- Opened all three files and checked what was inside each one.
- Fixed the messy category names.
- Removed rows that were missing important information.

Step 2: Combine the Data

- Joined the reactions file with the scoring file, so every reaction now carried its point value.
- Joined that result with the content file, so every reaction also knew its category and content type.
- Fixed the "User ID" mix-up described above.

Step 3: Build the Charts

- Added up the total score for each category, reaction type, and sentiment (positive/neutral/negative).
- Turned those totals into both simple charts and interactive, clickable charts.

Step 4: Look at Time Patterns

- Converted the plain-text dates into real dates.
- Pulled out the year, month, and hour from each date.
- Sorted each hour into a time of day: Morning, Afternoon, Evening, or Night.
- Charted how reactions changed across years, months, and times of day.

Project Iterations

Round 1

- Loaded, cleaned, and combined the three files. Added up the scores and found the top 5 categories, shown as a first chart.

Round 2

- Added more charts (sentiment, content type, and time patterns), built interactive versions of the key charts, and gave every chart a consistent, modern design.

Findings and Analysis

Which Categories Win? (Top 5)

The 5 categories with the most positive engagement were: animals, science, healthy eating, technology, and food. Animals came out on top, with 68,624 total points.

This same ranking was checked against a completely separate project that used the exact same underlying data -- and it produced the identical top 5, in the identical order. That match is a strong sign this result is correct, not a coincidence or a mistake.

Looking closer at the numbers shows a useful pattern: the top 5 categories are tightly clustered together (61,598 to 68,624 points, only about 11% apart), while the lower half of the list drops off more steeply (down to 45,751 for public speaking, a 33% gap from the top). This suggests the top 5 form a genuinely distinct, higher-performing tier of content -- not just a handful of categories that happened to score a little higher than the rest -- which makes them a much stronger, more confident basis for an investment decision.

Scale matters here too: together, these 5 categories account for 36% of ALL engagement earned across the entire platform -- meaningfully more than the 31% an even split across 16 categories would produce. In plain terms, roughly a third of everything Social Buzz users respond to positively is concentrated in just under a third of its content categories.

It is also worth understanding HOW these categories are winning. The average score earned per individual reaction is nearly identical across every category platform-wide -- about 39 to 41 points per reaction, a spread of under 4%. In other words, a single reaction to an animals post is not inherently more enthusiastic than a reaction to a public speaking post. Animals wins because it draws MORE total reactions, not because each reaction scores higher. Practically, this reframes the opportunity: growing these categories further is a reach/volume problem (getting more eyes on the content), not a content-quality problem (making each post more emotionally compelling) -- the two call for different strategies.

Top 5 Content Categories by Total Engagement Score

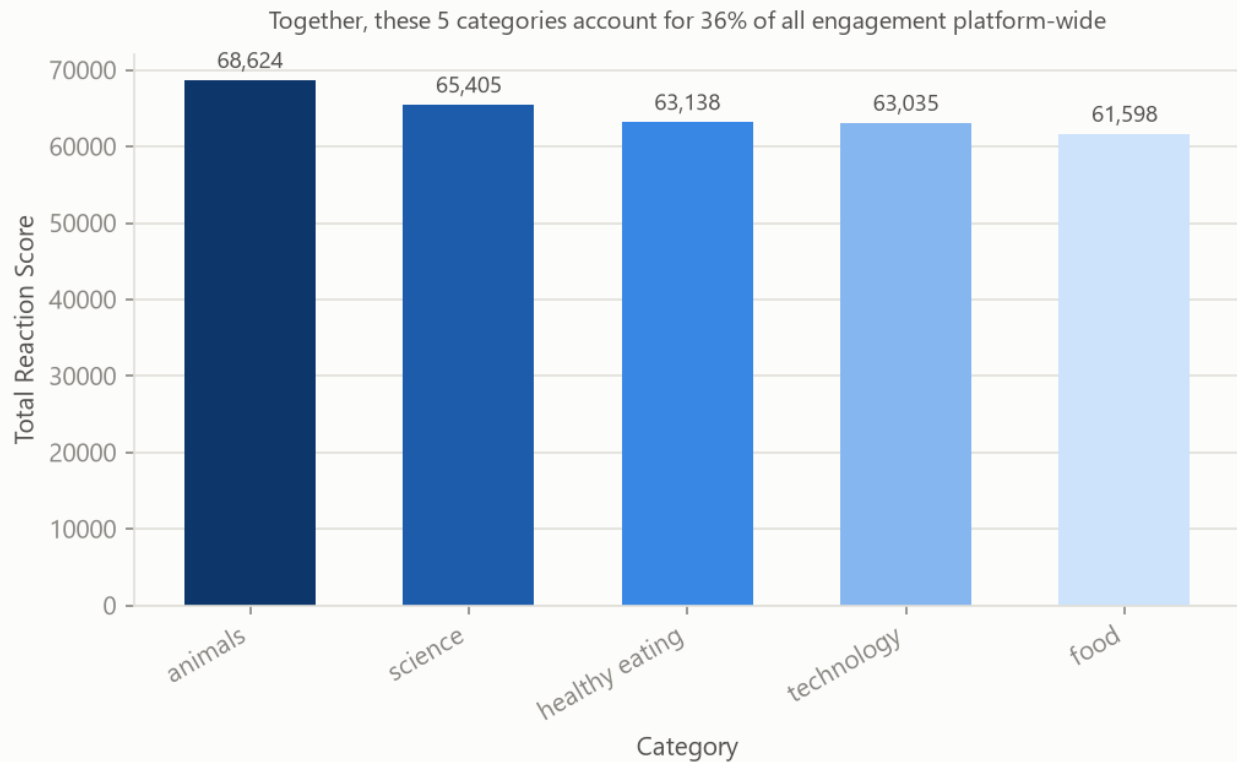


Figure 1: The top 5 content categories by total engagement score.

Key Takeaway: Animals leads by about 3,200 points over second-place science -- a clear front-runner, not a marginal one.

Sum of Score by Category

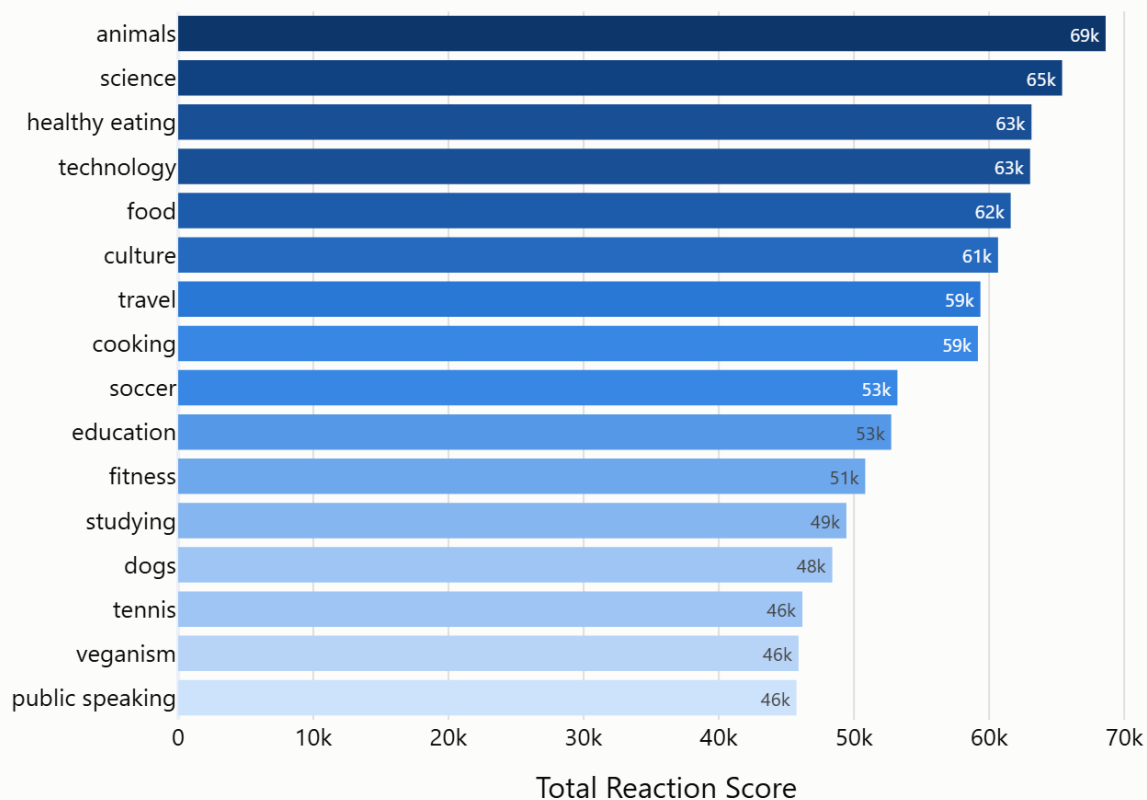


Figure 2: Total engagement score across all 16 categories, ranked highest to lowest.

Key Takeaway: After the top 5, scores decline gradually -- the bottom 8 categories all sit within about 7,000 points of each other, a much tighter spread than the top tier.

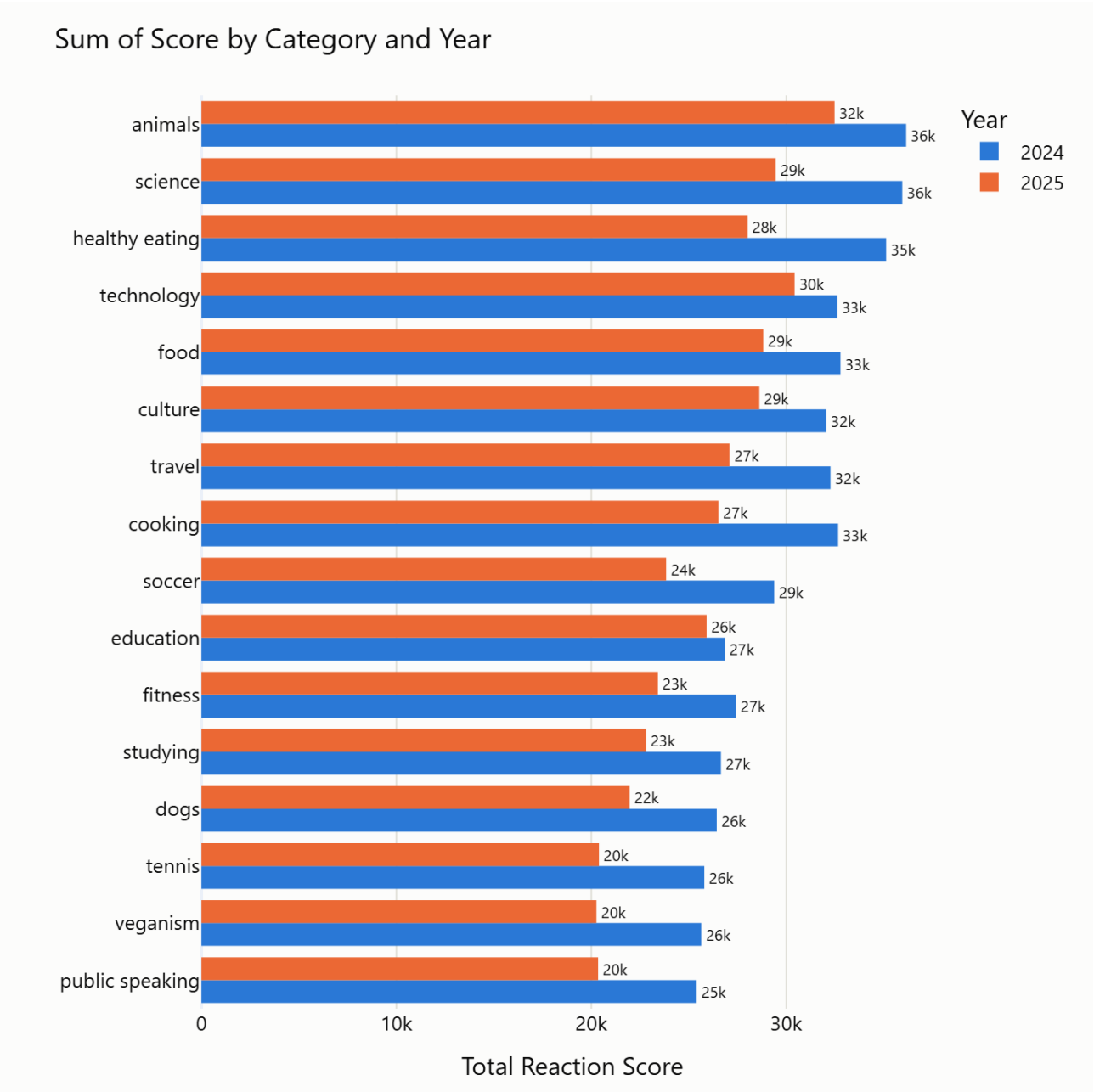


Figure 3: Total engagement score by category, comparing 2024 vs. 2025.

Key Takeaway: The category ranking barely changes between years -- animals and science lead in both 2024 and 2025.

What Kind of Content Gets Posted?

Posts were fairly evenly split across four content types: photo (about 27% of reactions), video (25%), GIF (25%), and audio (23%). No single type dominates -- Social Buzz users engage with all four in roughly equal amounts.

Because engagement barely differs by format, the format of a post (photo, video, GIF, or audio) does not appear to be a major driver of engagement on its own. The clear strategic takeaway: Social Buzz should prioritize WHAT topic to post about (the winning categories above) over WHICH format to post it in, since format alone shows little influence on the result.

One nuance worth flagging: WITHIN the top 5 categories specifically, audio content makes up a noticeably larger share of reactions (27%) than it does across the rest of the platform (21%), while video and GIF are both relatively less common there. This is a modest pattern, not a strong signal on its own -- but it is a reasonable hypothesis worth testing further before assuming photo or video is always the safer format choice for high-performing categories.

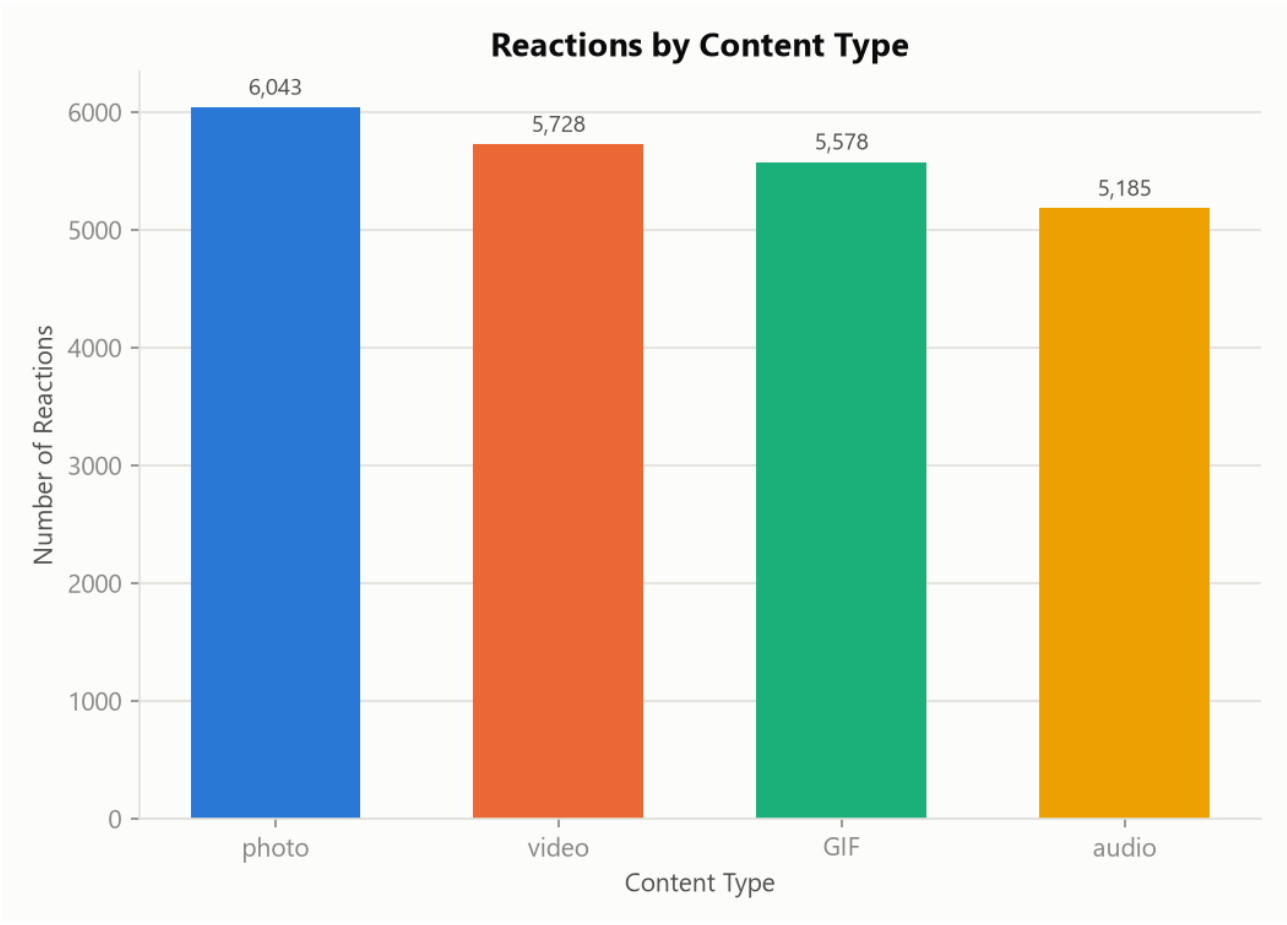


Figure 4: Number of reactions by content type.

Key Takeaway: Photo leads with about 6,000 reactions, but every content type stays within roughly 900 reactions of the top spot.

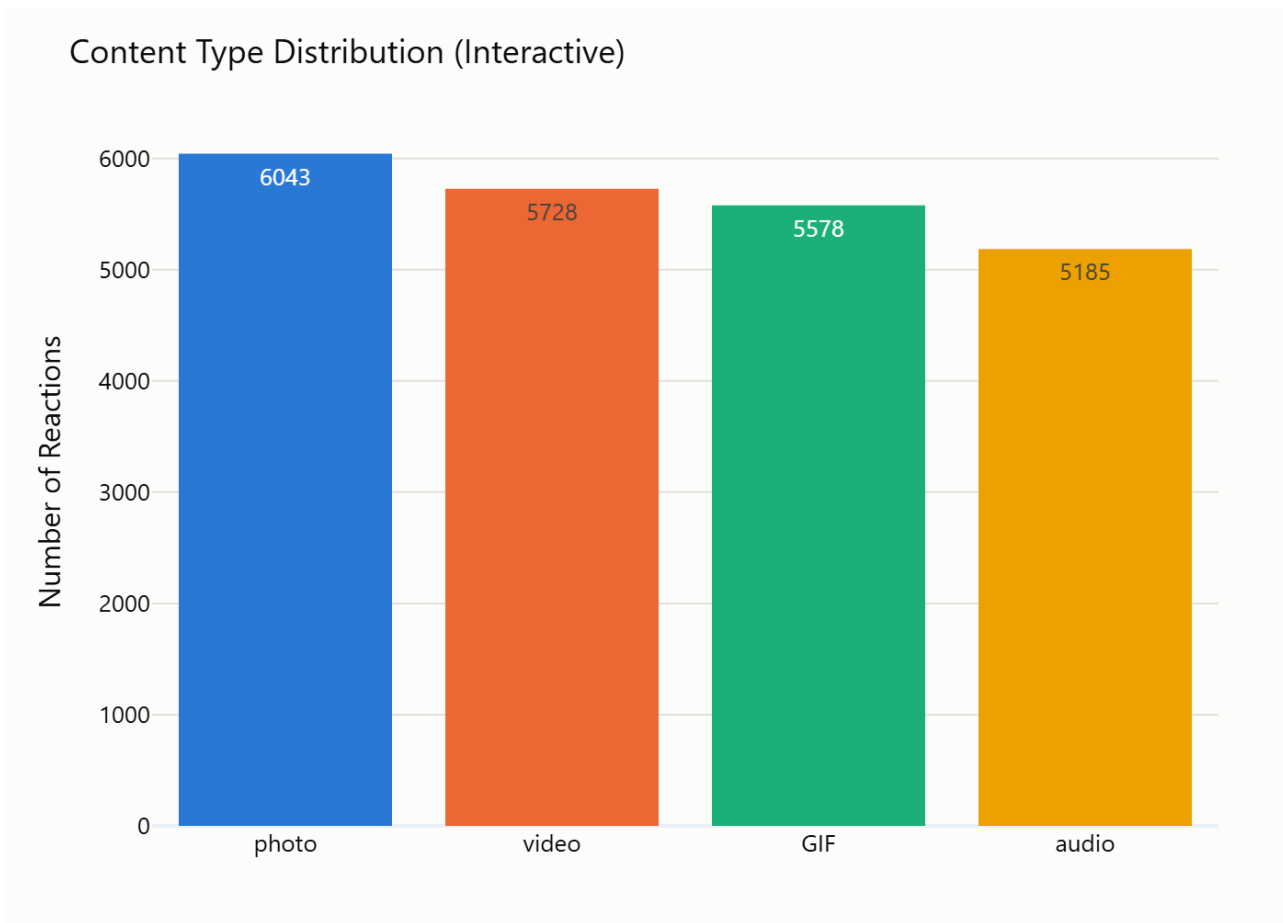


Figure 5: The same breakdown, as an interactive chart.

Key Takeaway: Same near-even split as Figure 4 -- open the matching .html file in the charts/ folder to hover for exact counts.

How Do People React?

The single reaction type that earned the most total points was "super love" (104,475 points), followed by "adore" and "want." At the other end, "disgust" earned almost no points, since it is worth 0 points by design.

Looking at all reactions together: 56% were positive, 31% were negative, and 12% were neutral. When counting actual points instead of just the number of reactions, positive reactions pull even further ahead, since positive reaction types tend to be worth more points each.

The fact that stronger reactions like "super love" and "adore" earn far more points than a plain "like" rewards content that inspires genuine enthusiasm, not just passive approval. It would be reasonable to guess the top 5 categories win by earning disproportionately MORE of these strong, positive reactions -- but checking that directly shows otherwise: the positive/neutral/negative sentiment mix in the top 5 categories (56.5% positive) is nearly identical to the rest of the platform (56.1% positive). This confirms the earlier finding -- the top 5 win on volume of reactions, not on a more positive or enthusiastic mix of reactions. At the same time, a 31% negative share platform-wide is not negligible: while overall sentiment is healthy, this is worth monitoring rather than dismissing outright.

Sum of Score by Reaction Type

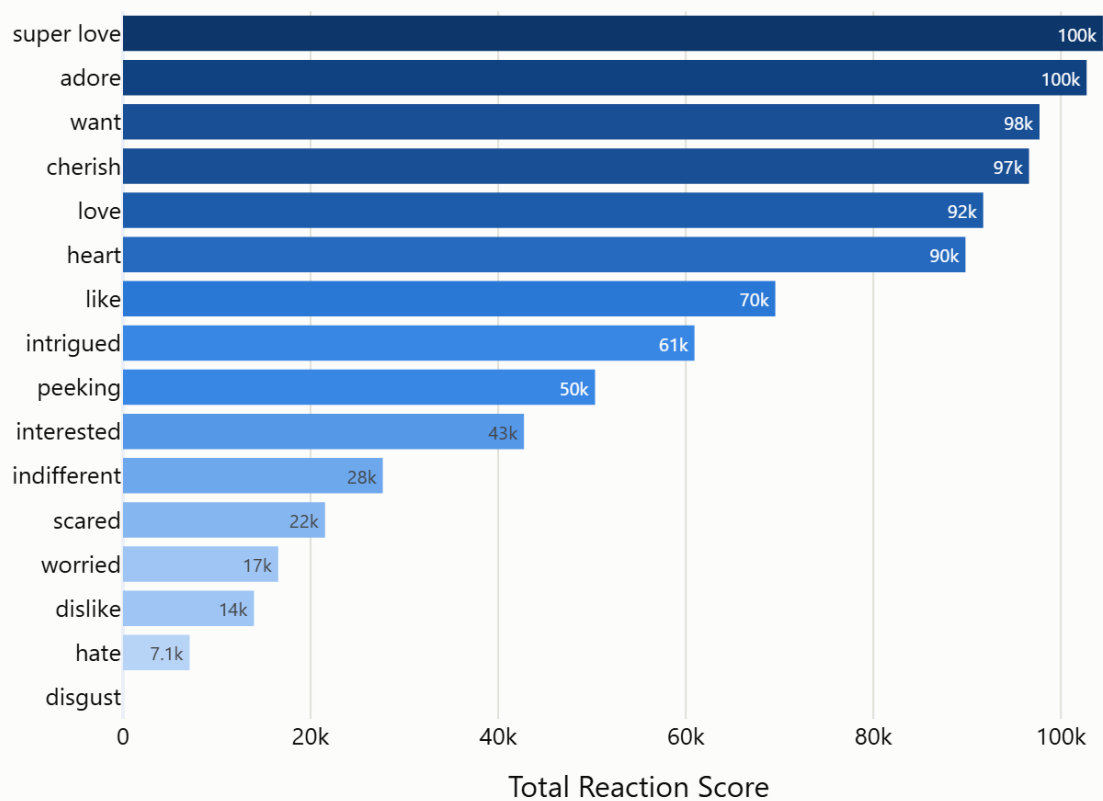


Figure 6: Total engagement score for each individual reaction type.

Key Takeaway: The top 3 reaction types alone (super love, adore, want) outscore the bottom 8 reaction types combined.

Overall Sentiment Breakdown of Reactions

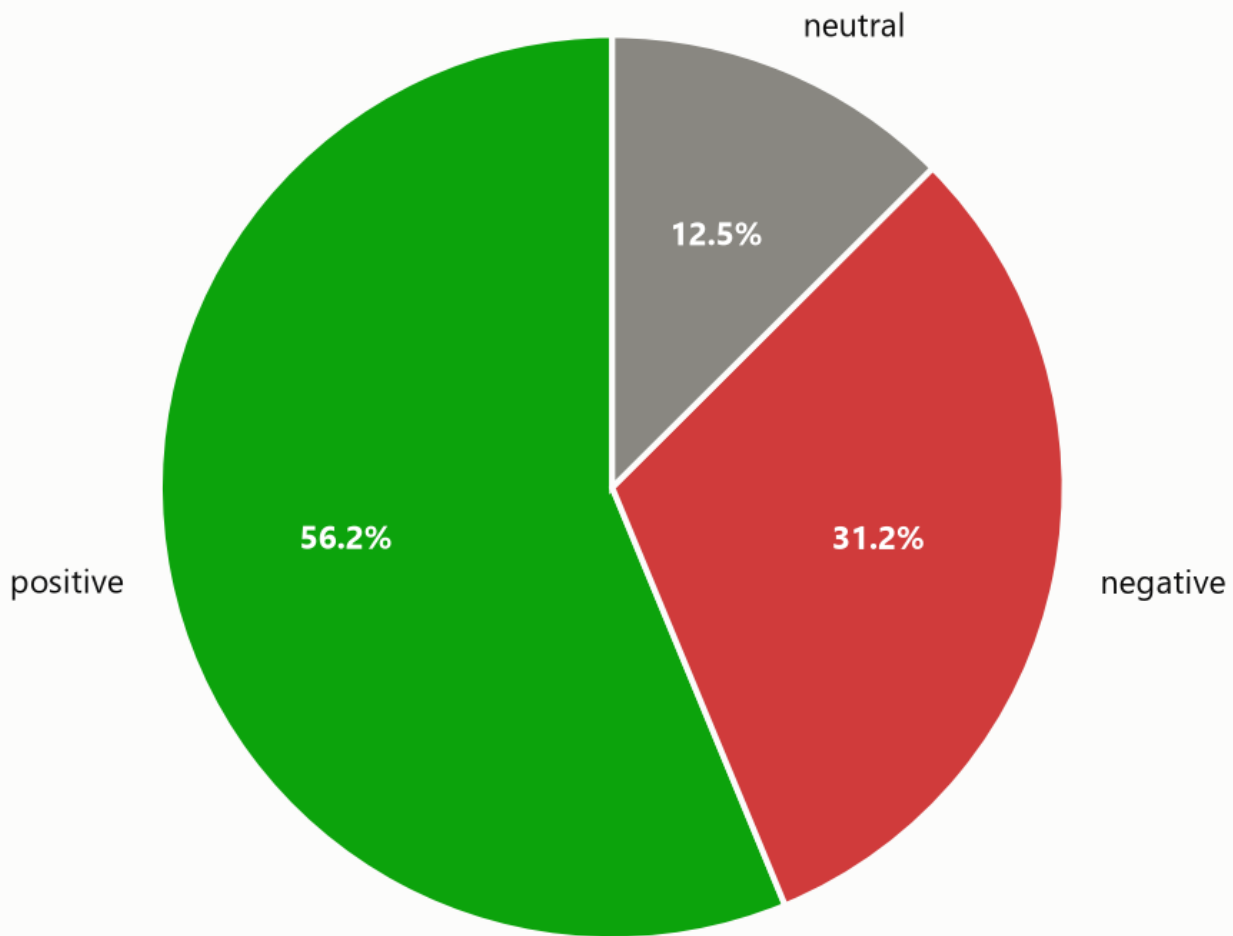


Figure 7: Overall sentiment breakdown -- what share of reactions were positive, neutral, or negative.

Key Takeaway: Positive reactions outnumber negative ones by nearly 2 to 1 (56.2% vs. 31.2%).

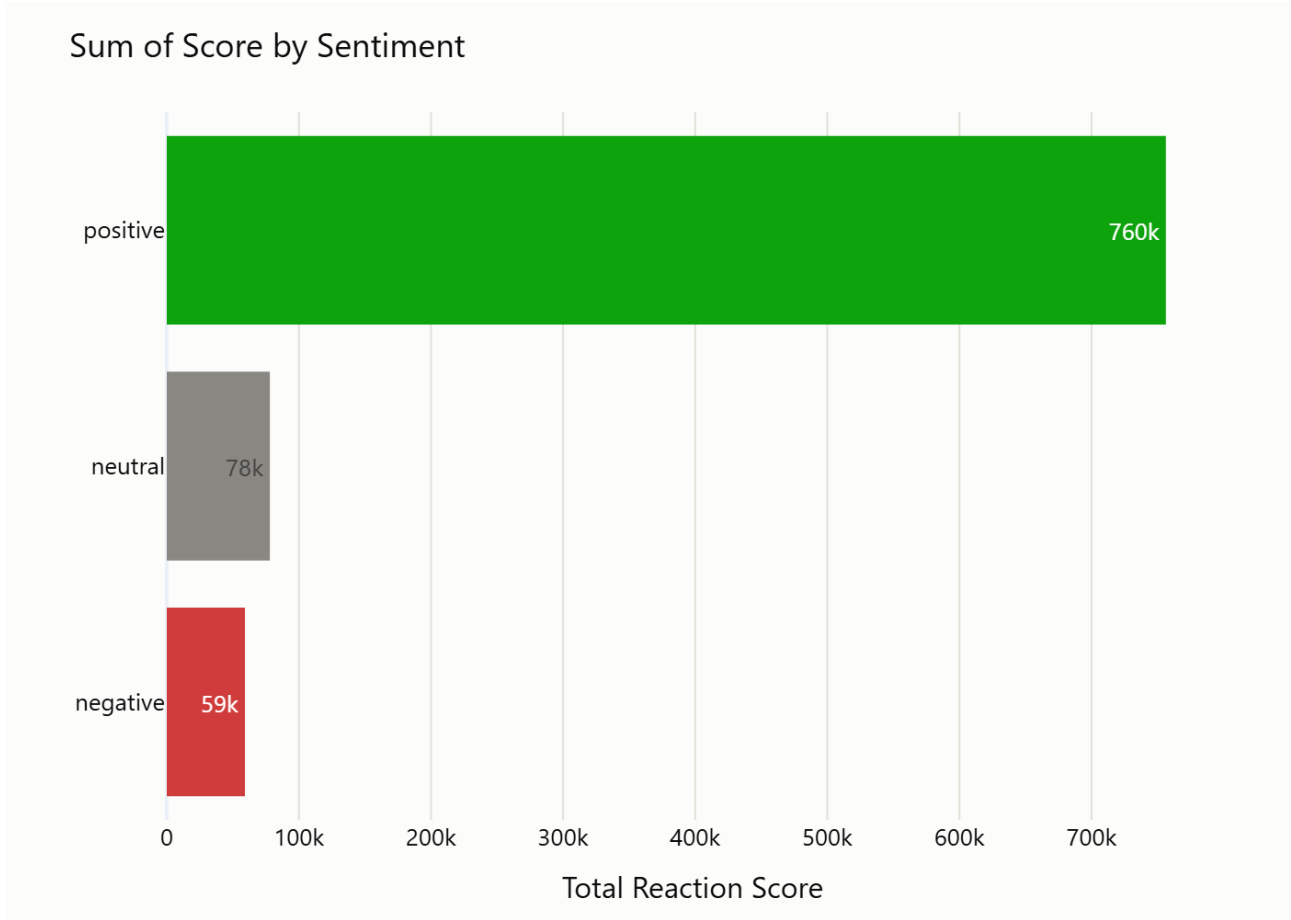


Figure 8: Total engagement score by sentiment.

Key Takeaway: Positive reactions account for about 85% of all points earned platform-wide (756,304 of 893,482 total).

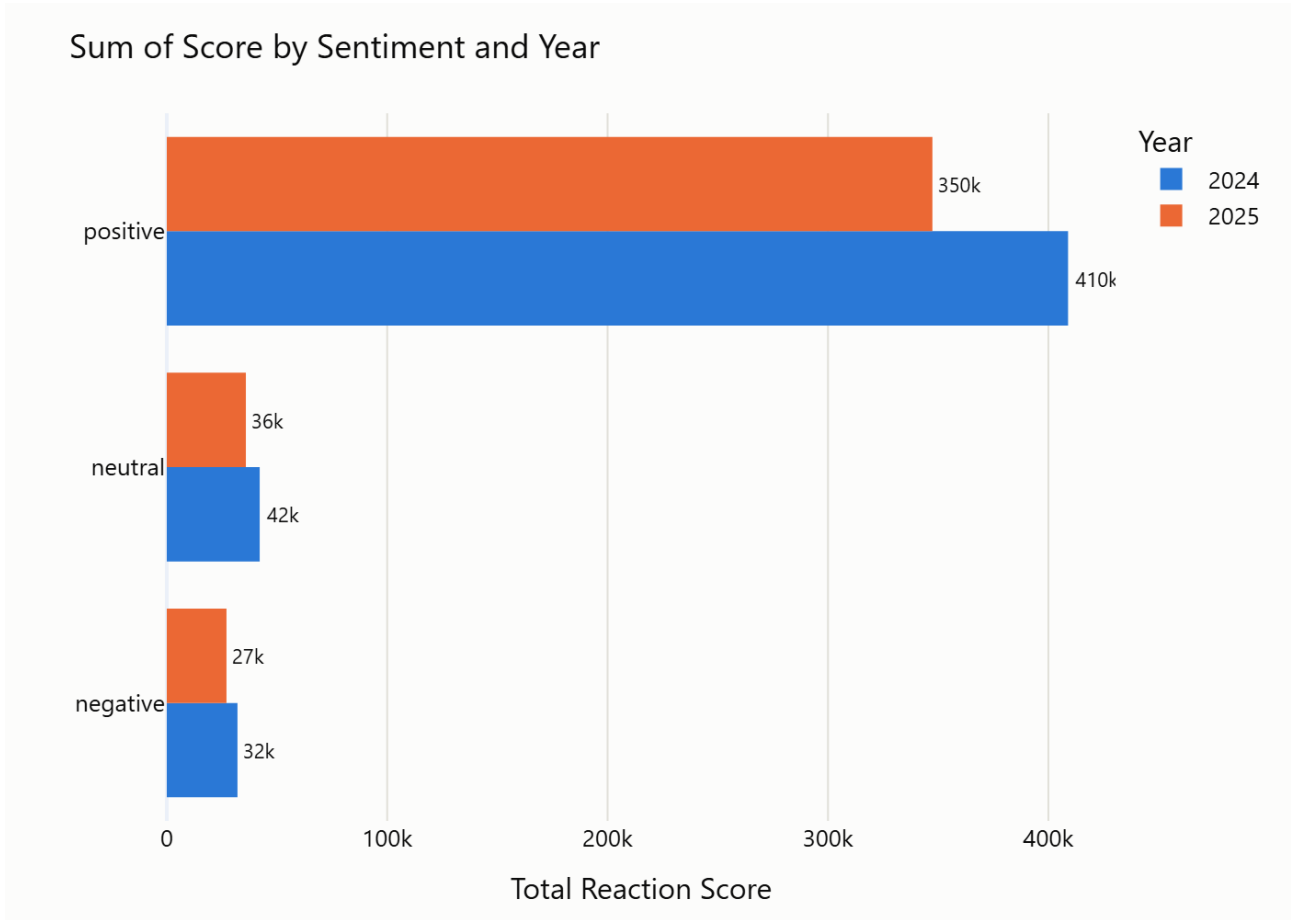


Figure 9: Total engagement score by sentiment, comparing 2024 vs. 2025.

Key Takeaway: The positive-dominant pattern holds in both years -- the sentiment mix barely shifted from 2024 to 2025.

Does Timing Matter?

At first glance, 2024 shows noticeably more reactions than 2025 (about 12,200 vs. 10,300). But this needs a closer look before drawing any conclusion from it: the dataset covers exactly one rolling year (mid-June to mid-June), not two complete calendar years -- so the 2024 label happens to cover about one extra month of data compared to 2025. Comparing only full months, the average reaction count per month is nearly identical between the two periods (about 1,898 vs. 1,862 per month). In other words, engagement did not meaningfully change year over year once the uneven date coverage is accounted for -- a useful reminder that a year-over-year chart can be misleading on its own without checking what date range each bar actually represents.

Month by month, activity stayed fairly steady all year, with May the busiest month and February the quietest -- no strong seasonal pattern. Time of day made little difference either, with mornings only slightly busier than afternoons, evenings, or nights. Practically, this means Social Buzz does not need to time posts around a specific window; a simple, consistent publishing schedule captures engagement just as well, freeing attention to focus on content category and quality instead.

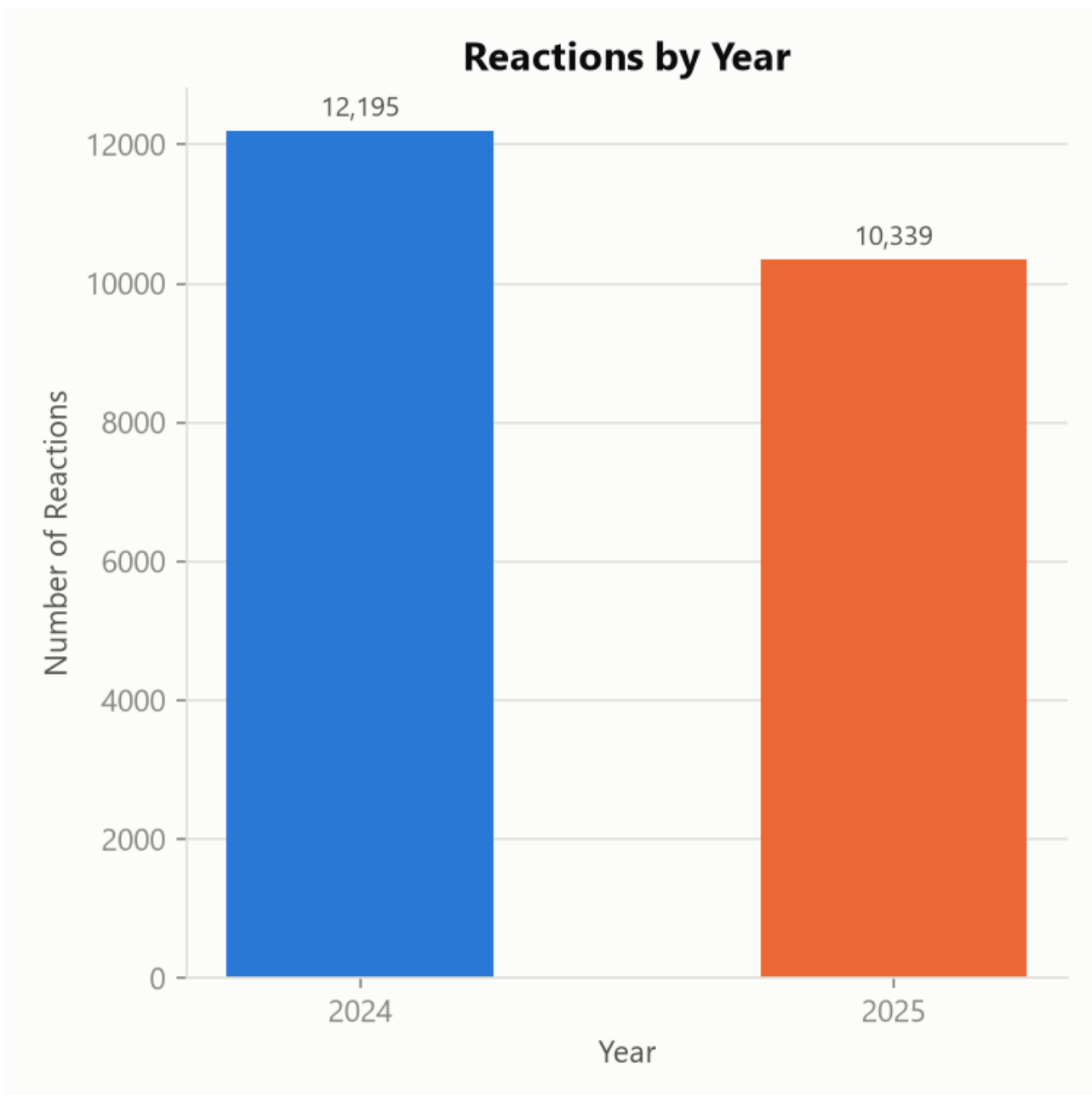


Figure 10: Number of reactions by year.

Key Takeaway: The 2024-vs-2025 gap shown here is a date-range artifact (see the paragraph above), not a real change in engagement.

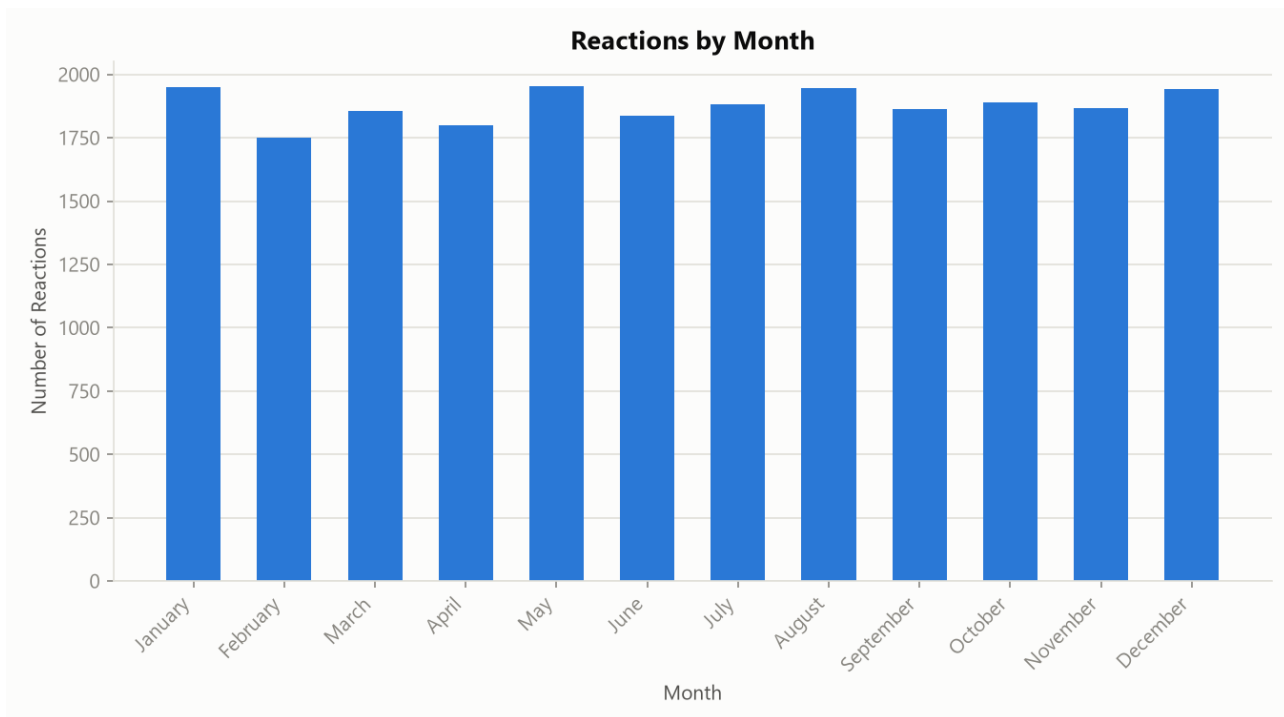


Figure 11: Number of reactions by month, in calendar order.

Key Takeaway: Monthly totals stay within about 200 reactions of each other all year -- no clear seasonal pattern.

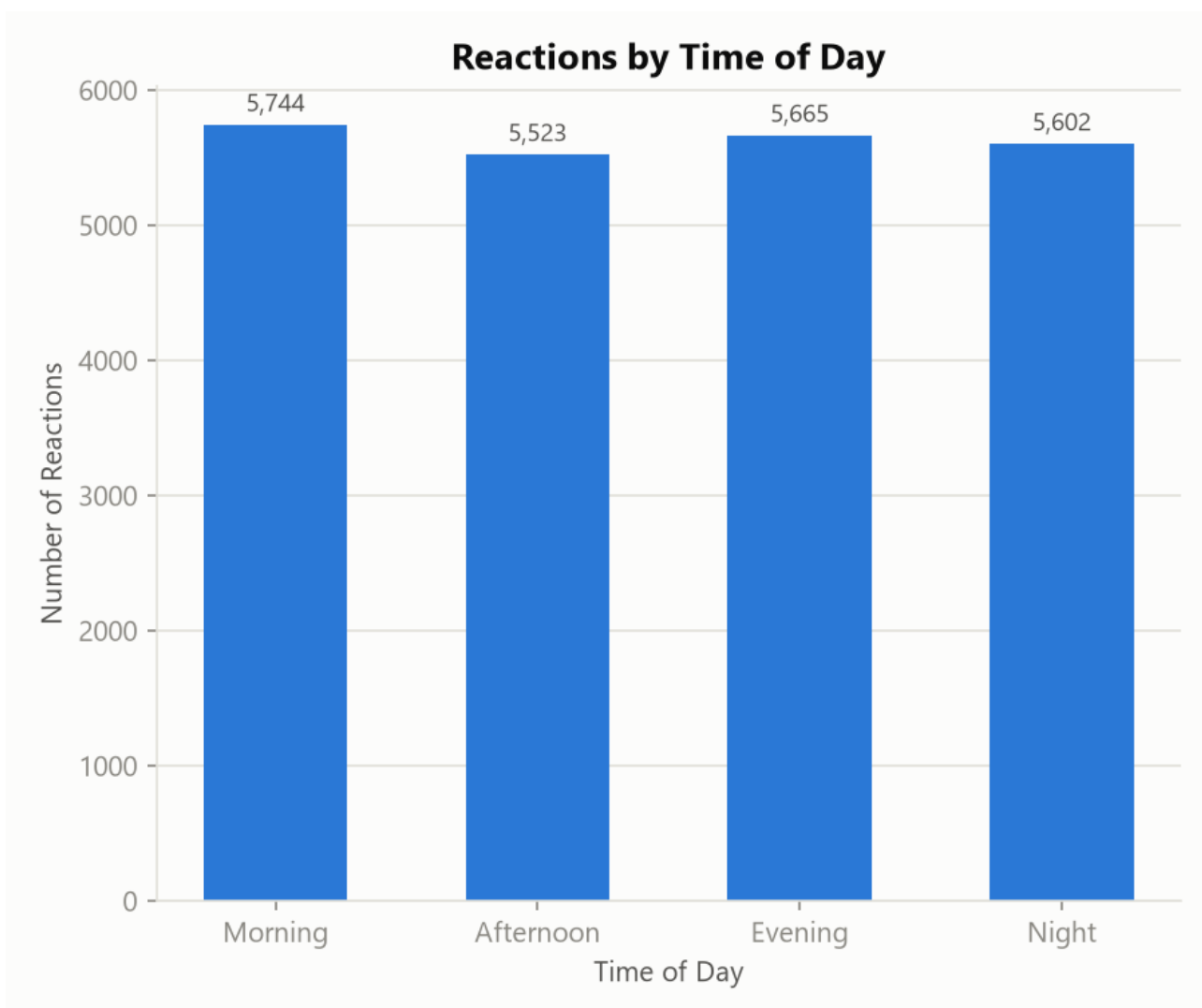


Figure 12: Number of reactions by time of day.

Key Takeaway: All four time windows stay within about 220 reactions of each other -- timing has little effect on

engagement.

Conclusion

This project took three messy, separate spreadsheets -- 1,000 posts and 25,553 raw reactions -- and turned them into one clean, unified table of 22,534 usable records. Along the way, several real data problems had to be fixed: missing information, inconsistent text, and a confusing date format.

The headline result: animals, science, healthy eating, technology, and food are the 5 content categories Social Buzz should invest in the most, since they earn the most positive reactions from users. This result was checked against a completely separate project using the same underlying data, and it matched exactly -- giving strong confidence that the finding is correct.

Learnings

- How to clean up real, messy data: missing values, inconsistent text, and confusing date formats.
- How to combine multiple related spreadsheets into one table, and resolve naming clashes along the way.
- How to build both simple charts and interactive, clickable charts, with a consistent visual design.
- How to work with dates and times in code, including writing a small custom function from scratch.
- How to diagnose and fix a real problem caused by the computer's own settings, not just the code.

Recommendations

- Invest more in animals, science, healthy eating, technology, and food content -- these are the categories users respond to most positively, together already earning 36% of all platform engagement.
- Because the top categories win on reaction VOLUME rather than reaction quality (the average score per reaction is nearly flat across every category), growing them further is a reach problem, not a content-quality problem -- prioritize distribution and discoverability (posting frequency, recommendation placement) for these categories over trying to make individual posts more "enthusiasm-worthy."
- Since most reactions (56%) are positive and only 31% are negative, the platform's overall mood is healthy -- but it's still worth keeping an eye on that negative third.
- Since reactions like "super love" and "adore" earn far more points than a plain "like," it may be worth encouraging users toward these stronger, more positive reactions.
- Content type and time of day don't meaningfully affect engagement right now, so no urgent change is needed on either front -- the one exception worth a closer look is the higher audio share within the top 5 categories specifically.